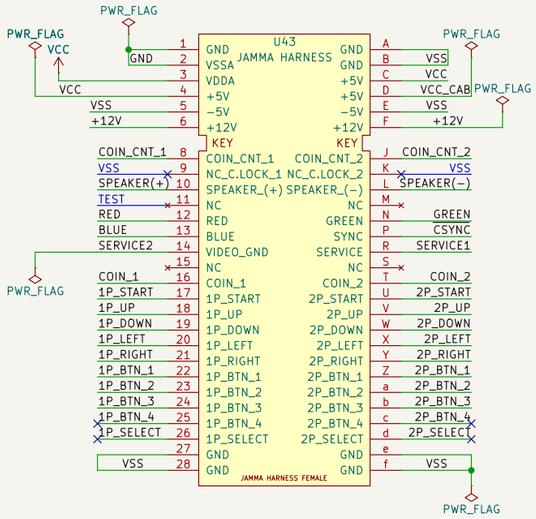
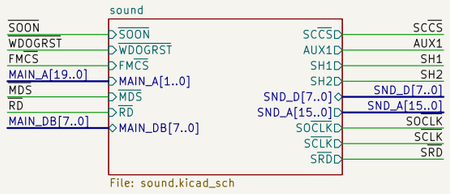
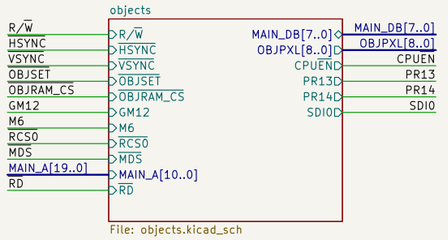
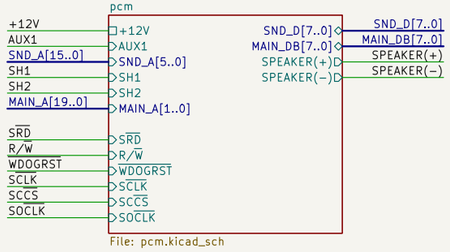
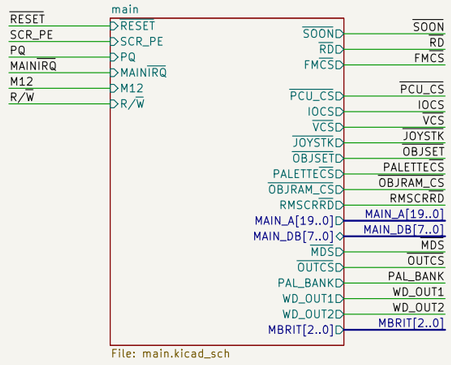
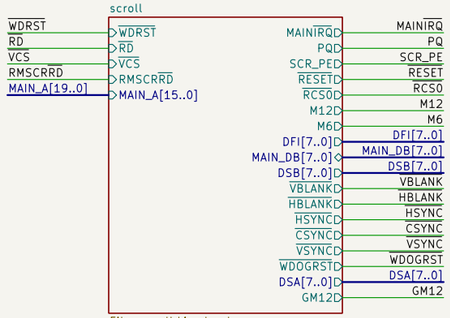


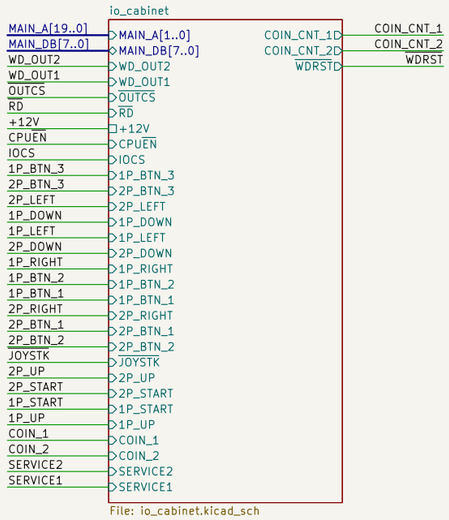


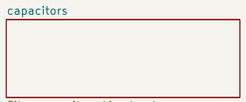


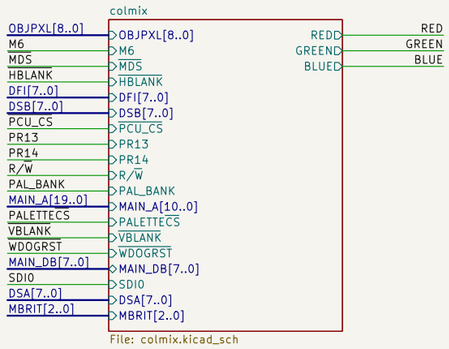












PWB352020B

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Sheet: /
File: surprise_attack.kicad_sch

Title: Surprise Attack

Size: A3	Date: 2025-09-01	**Rev: José Tejada G.**
KiCad E.D.A. 10.0.4		Id: 1/10

MAIN_D[7..0]

TEST HEADER
FOOTPRINT MAIN CPU

PAL2
PROG_A13= PAGED ? A15 : A13
PROG_A14= PAGED ? A16 : A14
PROG_A15= PAGED ? A17 : A15
PROG_A16= PAGED ? A18 : 0
PROGCS1 = A19 & PAGED
PROGCS2 = UNPAGED & (A19 | PAGED)

Floating (no driver)
verified on the PCB
while running the game
Constant 1.8V observed

PAL3
o12 = i7 = 2000-2FFF when BANK1
2800-5FFF
6000-7FFF when BANKR
o13 = i6 = 3C00-3FFF
PAGED= 6000-7FFF when BANKR
8000-9FFF
PALETTE_CS= 0000-07FF when WRK
OBJRAM_CS= 2000-27FF when BANK1
WORKCS= 0000-07FF when WRK
0800-1FFF
UNPAGED= A000-FFFF (ROM PPAL)

PAL1
JOYSTK 3F8C-3F8F
IO_CS 3F90-3F9F
MISC 3FC0-3FCF
PCU_CS 3FB0-3FBF
OBJSET 3FA0-3FAF
VCS= AS&i7&(A8&A7&A9&AB+i6+A7)

Connected to pin
17 of KONAMI 053260
(D8) in Parodius Da!

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Sheet: /main/
File: main.kicad_sch

Title: Surprise Attack

Size: A3 | Date: 2025-09-01 | Rev: José Tejada G.
KiCad E.D.A. 10.0.4 | Id: 2/10

RN4
SERVICE1
SERVICE2
COIN_1
COIN_2
KONAMI_005273
ISERVICE1
ISERVICE2
ICOIN_1
ICOIN_2
RN2
1P_START
2P_START
1P_UP
2P_UP
KONAMI_005273
I1P_START
I2P_START
I1P_UP
I2P_UP
RN3
1P_DOWN
2P_DOWN
1P_LEFT
2P_LEFT
KONAMI_005273
I1P_DOWN
I2P_DOWN
I1P_LEFT
I2P_LEFT
RN5
1P_RIGHT
2P_RIGHT
1P_BTN_1
2P_BTN_1
KONAMI_005273
I1P_RIGHT
I2P_RIGHT
I1P_BTN_1
I2P_BTN_1
RN1
1P_BTN_2
2P_BTN_2
1P_BTN_3
2P_BTN_3
KONAMI_005273
I1P_BTN_2
I2P_BTN_2
I1P_BTN_3
I2P_BTN_3
VCC
RA4
2228
2.2kΩX8
SW_DIPS_08
SW1
S15
S16
S17
VSS
C24
S10
ISERVICE1
I2P_START
I1P_START
JOYSTK
S11
ISERVICE2
I2P_UP
I1P_UP
JOYSTK
A0
A1
74LS253
MAIN_DB0
MAIN_DB1
C22
S12
ICOIN_1
I2P_DOWN
I1P_DOWN
JOYSTK
S13
ICOIN_2
I2P_LEFT
I1P_LEFT
JOYSTK
A0
A1
74LS253
MAIN_DB2
MAIN_DB3
VCC
RA2
2224
2.2kΩX4
SW_DIPS_04
SW3
VSS
C21
S14
I2P_RIGHT
I1P_RIGHT
JOYSTK
S15
I2P_BTN_1
I1P_BTN_1
JOYSTK
A0
A1
74LS253
MAIN_DB4
MAIN_DB5
MAIN_A[1..0]
MAIN_A0
MAIN_A1
74LS04
D21
S16
I2P_BTN_2
I1P_BTN_2
JOYSTK
S17
I2P_BTN_3
I1P_BTN_3
JOYSTK
A0
A1
74LS253
MAIN_DB6
MAIN_DB7
MAIN_DB[7..0]
RD
OUTCS
74LS32
F12B
VCC
EMI222M
C2
+12V
VSS
JP7
Jp2 Open
B17
CLK
VREF
RES
CT
VS
OUT1
OUT2
RESET
CNT2
CNT2
CNT1
CNT1
N.C
Master/RESET generation
for the whole system
and watchdog timer
KONAMI
051550
WDRST
COIN_CNT_2
COIN_CNT_1
WD_OUT1
WD_OUT2
USED IN THE GAME TMNT
Master /RESET generation for the
whole system and watchdog timer
coin counters handling.
C68
104
VSS
VCC
RA9
2228
2.2kΩX8
SW2
SW_DIPS_08
VSS
D22
CPUEN
VSS
IOCSD
74LS257
quad
2-1 MUX
MAIN_DB0
MAIN_DB1
MAIN_DB3
MAIN_DB2
D24
VCC
VSS
74LS257
quad
2-1 MUX
MAIN_DB4
MAIN_DB5
MAIN_DB7
MAIN_DB6
R13
470Ω
R14
100Ω
NOT POPULATED
CN8
VSS
SVCN4P

JOTEGO
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JOTEGO
Sheet: /io_cabinet/
File: io_cabinet.kicad_sch

Title: Surprise Attack

Size: A3	Date: 2025-09-01	**Rev: José Tejada G.**
KiCad E.D.A. 10.0.4		Id: 3/10

NOT POPULATED

RA8 4kX8 RA13 4kX8 RA12 4kX8 RA10 4kX8

GFXROMA[17..0]

K13 / K19 — 955005-06-07-08 KONAMI MASK 4M - 16BIT — CMOS ROM — HN62414P

GFXROMD[31..0]

J15 KONAMI 051962 TILE DRAWING

LAYER A COLOR INDEX · GFX ROM DATA · FIX LAYER COLOR INDEX · LAYER A TRANSPARENCY · FIX LAYER TRANSPARENCY · CPU DATA TO READ GFX ROMs · LAYER B TRANSPARENCY · LAYER B COLOR INDEX · CPU ADDRESS · MAIN CLK · LAYER A H SCROLL · LAYER B H SCROLL · READ GFX ROM ENABLE · SOUND · ATTRIBUTE CODE FROM 052109

DSA[7..0] SCRA

FIX DFI[7..0]

MAIN_DB[7..0]

DSB[7..0] SCRB

G22 XTAL_OSC 24MHz · CP1 104 · 74LS04 H21A · 74LS04 H21C

H22B 74LS74 · WDRST · WDOGRST

J20 KONAMI 052109 TILE MAP GENERATOR

30 pins per side

LAYER A SCROLL X · LAYER B SCROLL X · TILE ATTRIBUTES · GFX ROM ADDRESS · VRAM ADDRESS · GFX ROM read · VRAM DATA · CPU DATA

MAIN_A[15..0]

SCRAMA[12..0] · H24 · VD[15..0]

J24

MB8464A-10W-15W CMOS Static RAM 8K x 8 BITS

R18 470Ω · R20 100Ω · C15C 74LS07 · CSYNC

JP6 Jp_2_Bridged · JP5 Jp_2_Bridged

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JOTEGO

Sheet: /scroll/
File: scroll.kicad_sch

Title: Surprise Attack

Size: A3	Date: 2025-09-01	**Rev: José Tejada G.**
KiCad E.D.A. 10.0.4		Id: 4/10

NOT POPULATED

MAIN_A[10..0]

OBJROMA[18..0]

OBJROMD[31..0]

ORAMA[10..0]

OBJRAMD[7..0]

MAIN_DB[7..0]

HP[9..0]

AR[2..0]

PL[6..0]

BR[2..0]

OBJPXL[8..0]

ROM's MASK OF THE SPRITES

U41
KONAMI
053245
OBJECT GENERATOR

U42
053244

K2
955D05-06-07-08
KONAMI
MASK 4M - 16BIT
HN62414P
CMOS ROM

K8
955D05-06-07-08
KONAMI
MASK 4M - 16BIT
HN62414P
CMOS ROM

HM6116P-3
2kB RAM

H14
buffers
74LS245

F12C 74LS32
F13D 74LS32
F11D 74LS08
74LS32 F13C

Conn_02x32_Odd_Even_First

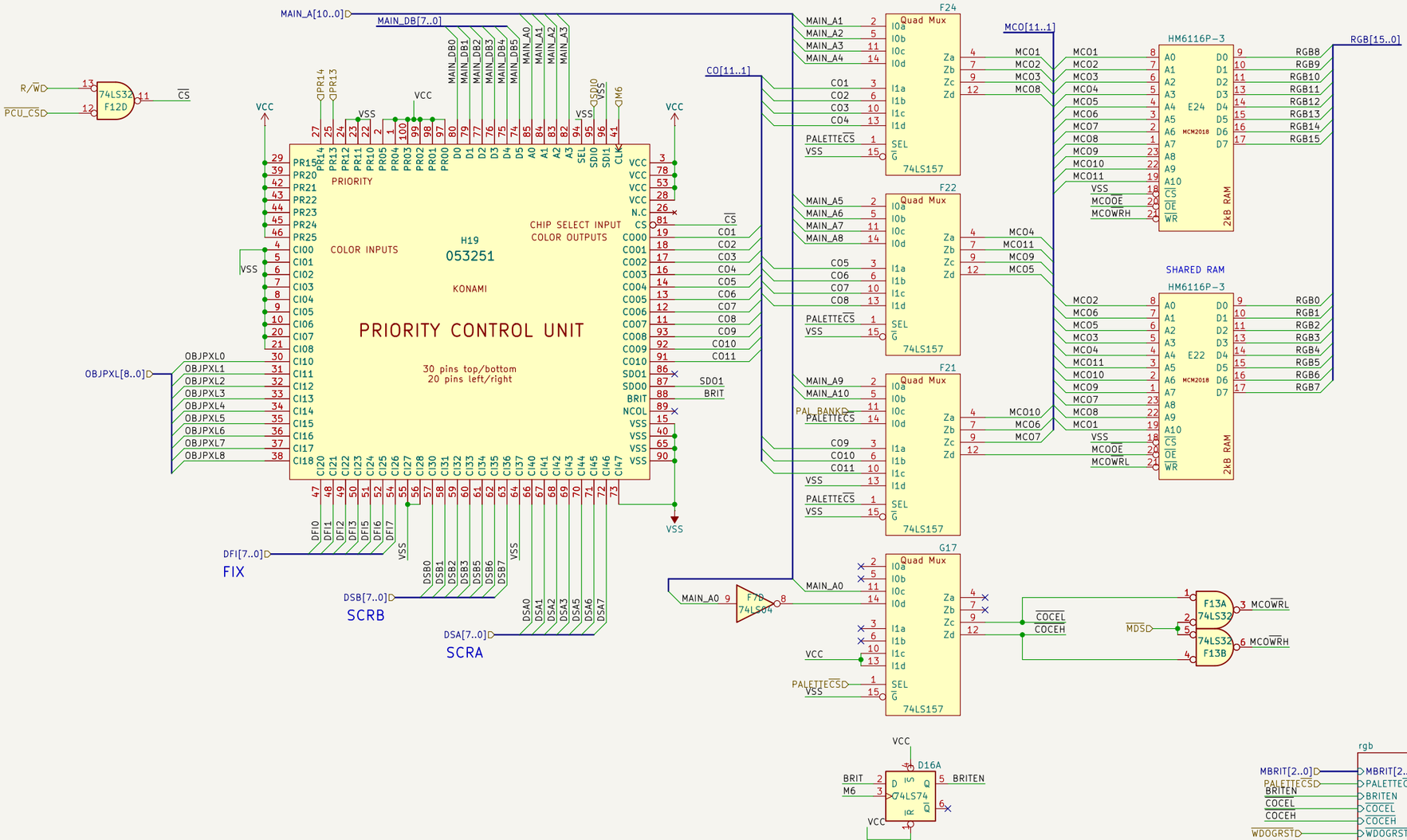
RA1 4kX8
RA11 4kX8
RA5 4kX8
RA3 4kX8

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Sheet: /objects/
File: objects.kicad_sch

Title: Surprise Attack

Size: A3	Date: 2025-09-01	**Rev: José Tejada**
KiCad E.D.A. 10.0.4		Id: 5/10



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Sheet: /colmix/
File: colmix.kicad_sch

Title: Surprise Attack

Size: A3	Date: 2025-09-01	**Rev: José Tejada G.**
KiCad E.D.A. 10.0.4		Id: 6/10

RGB[15..0]

MAIN_DB[7..0]

E21

MAIN_DB0 2 — A0 — B0 18 — RGB0
MAIN_DB1 3 — A1 — B1 17 — RGB1
MAIN_DB2 4 — A2 — B2 16 — RGB2
MAIN_DB3 5 — A3 — B3 15 — RGB3
MAIN_DB4 6 — A4 — B4 14 — RGB4
MAIN_DB5 7 — A5 — B5 13 — RGB5
MAIN_DB6 8 — A6 — B6 12 — RGB6
MAIN_DB7 9 — A7 — B7 11 — RGB7

buffers

PALETTECS 1 — A->B
COCELD 19 — CE

74LS245

RGB0 3 — D0 — Q0 2 — G3
RGB1 4 — D1 — Q1 5 — G4
RGB2 7 — D2 — Q2 6 — B0
RGB3 8 — D3 — Q3 9 — B1
RGB4 13 — D4 — Q4 12 — B2
RGB5 14 — D5 — Q5 15 — B3
RGB6 17 — D6 — Q6 16 — B4
18 — D7 — Q7 19 — SHADOW

D19

SDO1D 5 — C17F 74LS04 — 6 18

M6D 11 — Cp
VCC 1 — Mr

74LS273

flip flops

E20

MAIN_DB0 2 — A0 — B0 18 — RGB8
MAIN_DB1 3 — A1 — B1 17 — RGB9
MAIN_DB2 4 — A2 — B2 16 — RGB10
MAIN_DB3 5 — A3 — B3 15 — RGB11
MAIN_DB4 6 — A4 — B4 14 — RGB12
MAIN_DB5 7 — A5 — B5 13 — RGB13
MAIN_DB6 8 — A6 — B6 12 — RGB14
MAIN_DB7 9 — A7 — B7 11 — RGB15

buffers

PALETTECS 1 — A->B
COCEH 19 — CE

74LS245

RGB8 3 — D0 — Q0 2 — R0
RGB9 4 — D1 — Q1 5 — R1
RGB10 7 — D2 — Q2 6 — R2
RGB11 8 — D3 — Q3 9 — R3
RGB12 13 — D4 — Q4 12 — R4
RGB13 14 — D5 — Q5 15 — G0
RGB14 17 — D6 — Q6 16 — G1
RGB15 18 — D7 — Q7 19 — G2

D20

M6 11 — Cp
VCC 1 — Mr

74LS273

flip flops

F20

74LS164
High Speed 8-Bit Serial In Parallel-Out Shift Register

VBLANK 1 — A
HBLANK 2 — B AND
M6 8 — CLK
WDOGRST 9 — MR

Q0 3
Q1 4
Q2 5
Q3 6
Q4 10
Q5 11
Q6 12
Q7 13

C15A 74LS07 1 — 2
C15B 74LS07 3 — 4

G4 3 — C17B 74LS07 — 4
G3 5 — C19C 74LS07 — 6
G2 13 — C20F 74LS07 — 12
G1 3 — C19B 74LS07 — 4
G0 9 — C17D 74LS07 — 8

USED IN THE GAME LETHAL ENFORCERS

VCC
S2
Pull Up*
D4 2
D3* 3
D2* 4
D1* 5
D0* 6
BK* 7
DIGITAL INPUT ANALOG OUTPUT
AO 8
RGB
KONAMI 052535 5-bit Video DAC
VSS

GREEN

SHADOW 3 — C16B 74LS07 — 4
MBRIT2 13 — C17F 74LS07 — 12

NOT POPULATED
R21 N.M
R16 N.M

MBRIT1 13 — C19F 74LS07 — 12
MBRIT0 11 — E19E 74LS07 — 10

NOT POPULATED
R25 N.M
R15 N.M

B4 11 — C15E 74LS07 — 10
B3 9 — C15D 74LS07 — 8
B2 9 — C16D 74LS07 — 8
B1 11 — C16E 74LS07 — 10
B0 5 — C17C 74LS07 — 6

VCC
S1
Pull Up*
D4 2
D3* 3
D2* 4
D1* 5
D0* 6
BK* 7
DIGITAL INPUT ANALOG OUTPUT
AO 8
RGB
KONAMI 052535 5-bit Video DAC

BLUE

SHADOW 13 — C16E 74LS07 — 12
MBRIT2 11 — C17E 74LS07 — 10

NOT POPULATED
R23 N.M
R26 N.M

MBRIT1 13 — 74LS00 E16D — 11 MBRIT1 — C19E 74LS07 — 10
MBRIT0 12 5 — 74LS00 E16B — 6 MBRIT0 — E19A 74LS07 — 2
BRITEND 4 1 — E16A 74LS00 — 3 MBRIT2
MBRIT2 2

NOT POPULATED
R24 N.M
R22 N.M

MBRIT[2..0]

R4 11 — C20E 74LS07 — 10
R3 3 — C20B 74LS07 — 4
R2 5 — C20C 74LS07 — 6
R1 1 — C20A 74LS07 — 2
R0 9 — C20D 74LS07 — 8

SHADOW 1 — C16A 74LS07 — 2

NOT POPULATED
R12 118Ω
R27 N.M

MBRIT2 1 — C17A 74LS07 — 2

VCC
S3
Pull Up*
D4 2
D3* 3
D2* 4
D1* 5
D0* 6
BK* 7
DIGITAL INPUT ANALOG OUTPUT
AO 8
RGB
KONAMI 052535 5-bit Video DAC

RED

MBRIT1 1 — C19A 74LS07 — 2

NOT POPULATED
R19 N.M
R17 N.M

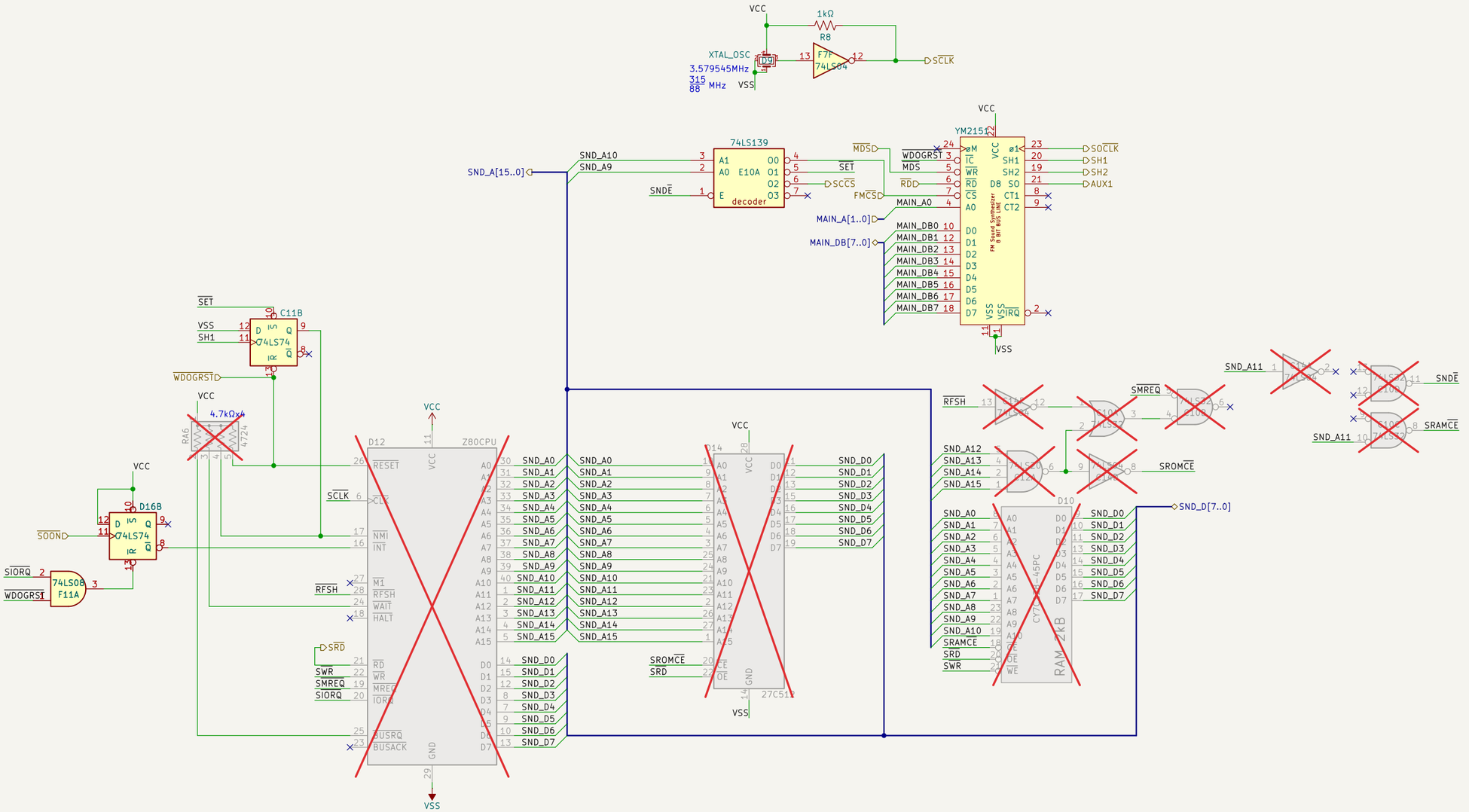
MBRIT0 13 — E19F 74LS07 — 12

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Sheet: /colmix/rgb/
File: rgb.kicad_sch

Title: Surprise Attack

Size: A3	Date: 2025-09-01	**Rev: José Tejada G.**
KiCad E.D.A. 10.0.4		Id: 7/10



JOTEGO
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Sheet: /sound/
File: sound.kicad_sch

Title: Surprise Attack

Size: A3	Date: 2025-09-01	**Rev: José Tejada G.**
KiCad E.D.A. 10.0.4		Id: 8/10

NOT POPULATED

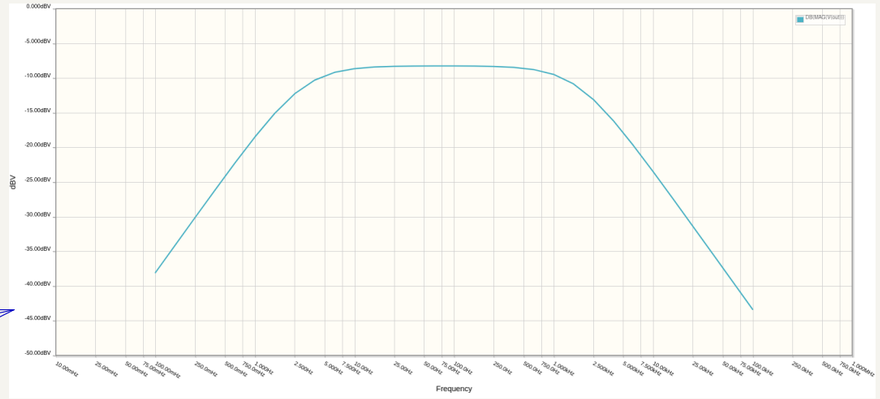
RA[19..0]

Communication between the main CPU and the audio CPU.

RD[7..0]

Connected to pin 21 of alternate YM2151 footprint.

Connected to pin 7 of 74LS139 (E10) in Parodius Da!

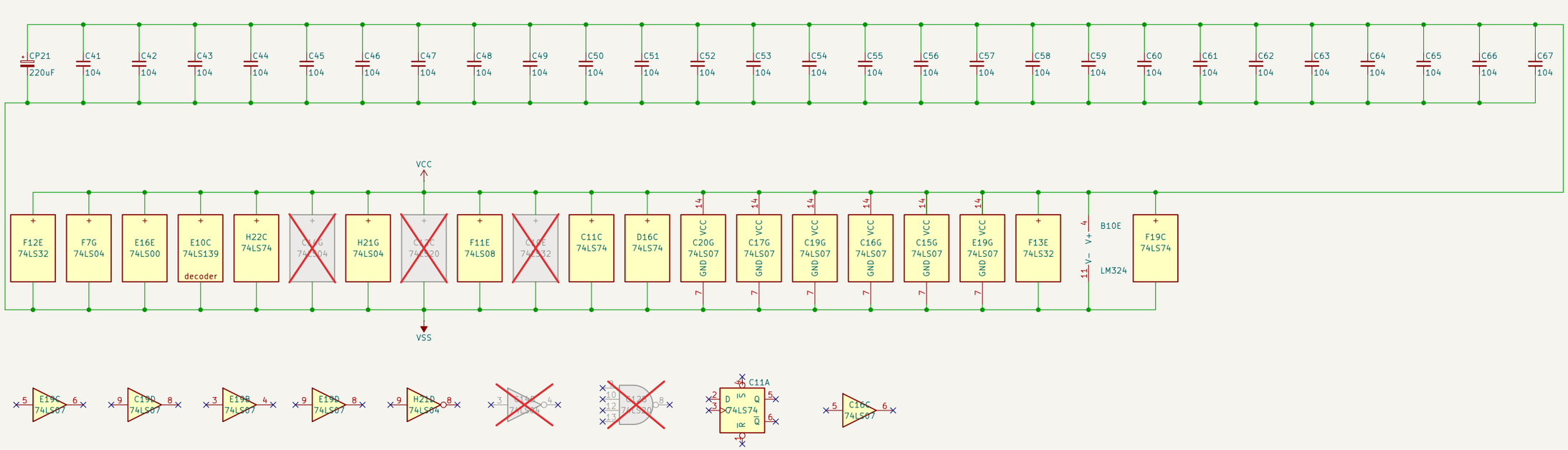


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Sheet: /pcm/
File: pcm.kicad_sch

Title: Surprise Attack

Size: A3	Date: 2025-09-01	**Rev: José Tejada G.**
KiCad E.D.A. 10.0.4		Id: 9/10



JOTEGO
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Sheet: /capacitors/
File: capacitors.kicad_sch

Title: Surprise Attack

Size: A3	Date: 2025-09-01	**Rev: José Tejada G.**
KiCad E.D.A. 10.0.4		Id: 10/10